Cover letter

Dear Professor Haueisen,

Please consider our Research Article, “Disentangling spatial-filter and classifier contributions to limited-label motor-imagery EEG recalibration”, for Biomedical Engineering / Biomedizinische Technik.

The manuscript addresses how a short calibration run should change a motor-imagery decoder. By giving classifier-only recalibration, full spatial refitting and support-only alignment the same source and current-session labels, we quantify the contribution of each update operation on identical query trials. In the original twelve-participant Kumar cohort, spatial refitting added 5.47 balanced-accuracy points beyond classifier recalibration and accounted for 73.7% of the total mean improvement over freezing. Prediction transitions and five-session participant summaries explain that increment.

The completed calibration-budget analysis adds a second spatial operation. Support-only alignment increased balanced accuracy by 5.30 points beyond full refitting in Kumar (pointwise 95% interval 2.81–8.26) and by 5.65 points in the separately fixed nine-participant BNCI comparison (3.15–8.70; secondary Holm $p=0.01172$). All nine external participants improved. Full versus classifier-only refitting in BNCI differed by 0.19 points (-1.85 to 2.13). Complete budget curves, marker-aware window sensitivity and the full neural/geometric controls accompany these comparisons.

The contribution is a reproducible analysis of where limited-label recalibration gains arise, supported by shared trial identities and participant-level estimates. The study uses our recorded analyses of public datasets; source methods and acquisition studies receive their appropriate citations. Data/code availability, national-law secondary-use ethics and actual AI assistance are documented in the separate declarations.

The manuscript has been reviewed and approved for submission by Ziyuan Zhu, Sihao Liu and Haoyang Yu, all sharing the Chongqing Medical University affiliation. All authors accept responsibility for its content and report no conflicts of interest and no study funding. The prior PLOS ONE submission was rejected, as confirmed by the corresponding author. The accompanying list supplies five suggested reviewers with institutional affiliations and professional contact details. The corresponding author confirms no known relevant conflicts between these candidates and any of the three authors.

Sincerely,

Ziyuan Zhu, corresponding author

College of Artificial Intelligence Medicine, Chongqing Medical University, Jinyun Campus, No. 61 Daxuecheng Middle Road, Shapingba District, Chongqing 401331, China

zzy2630816871@gmail.com

ORCID: https://orcid.org/0009-0005-1153-4926